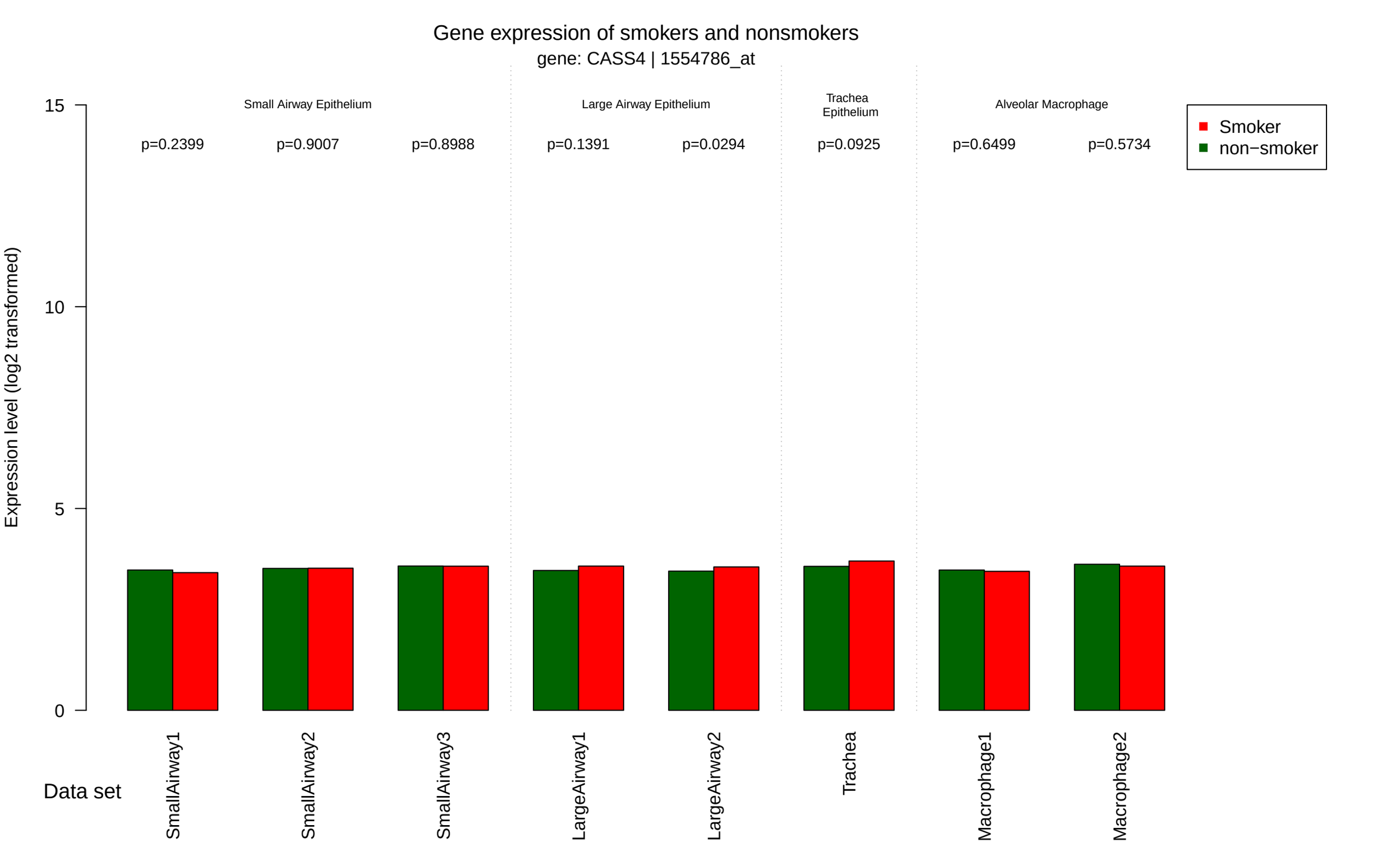
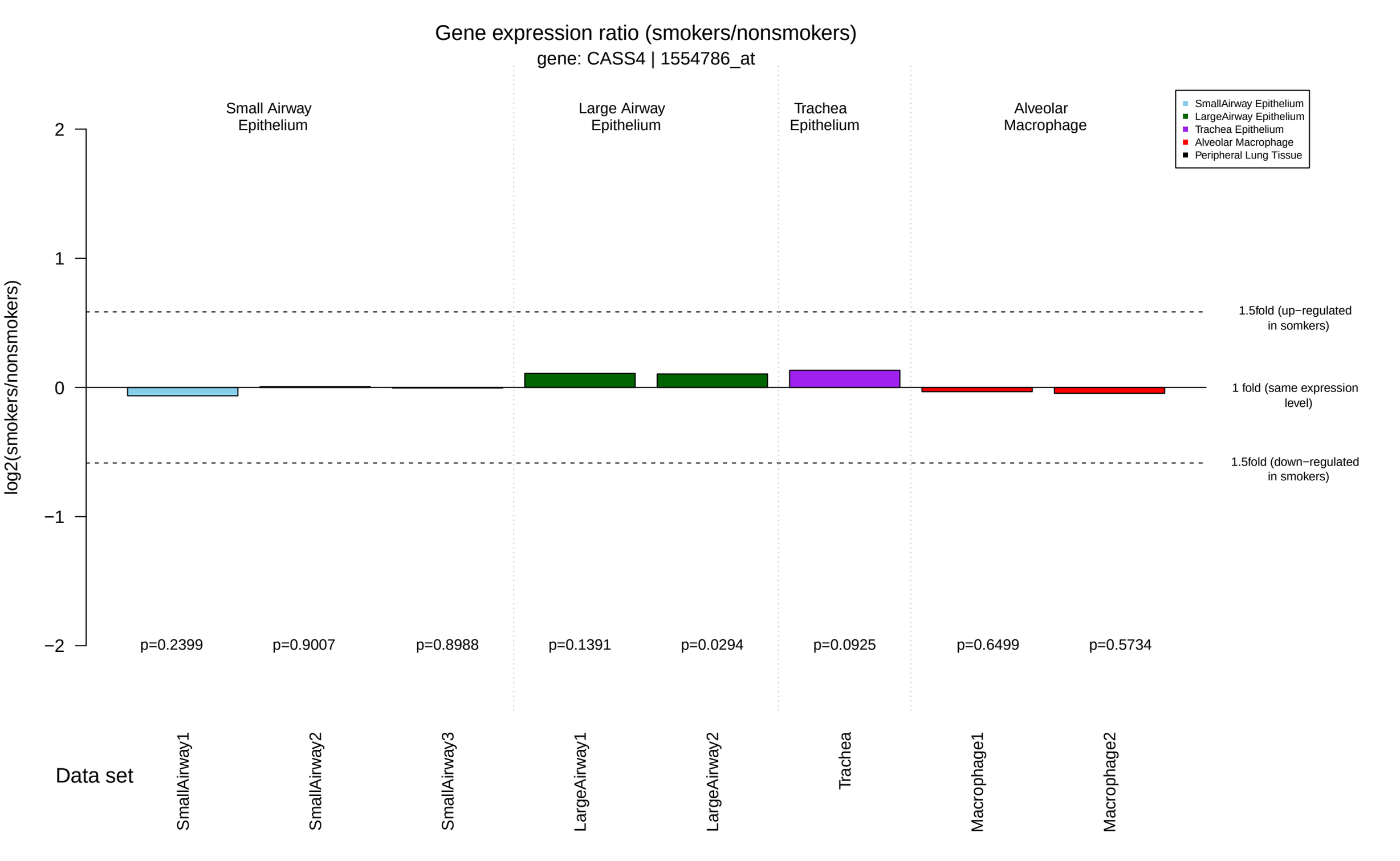
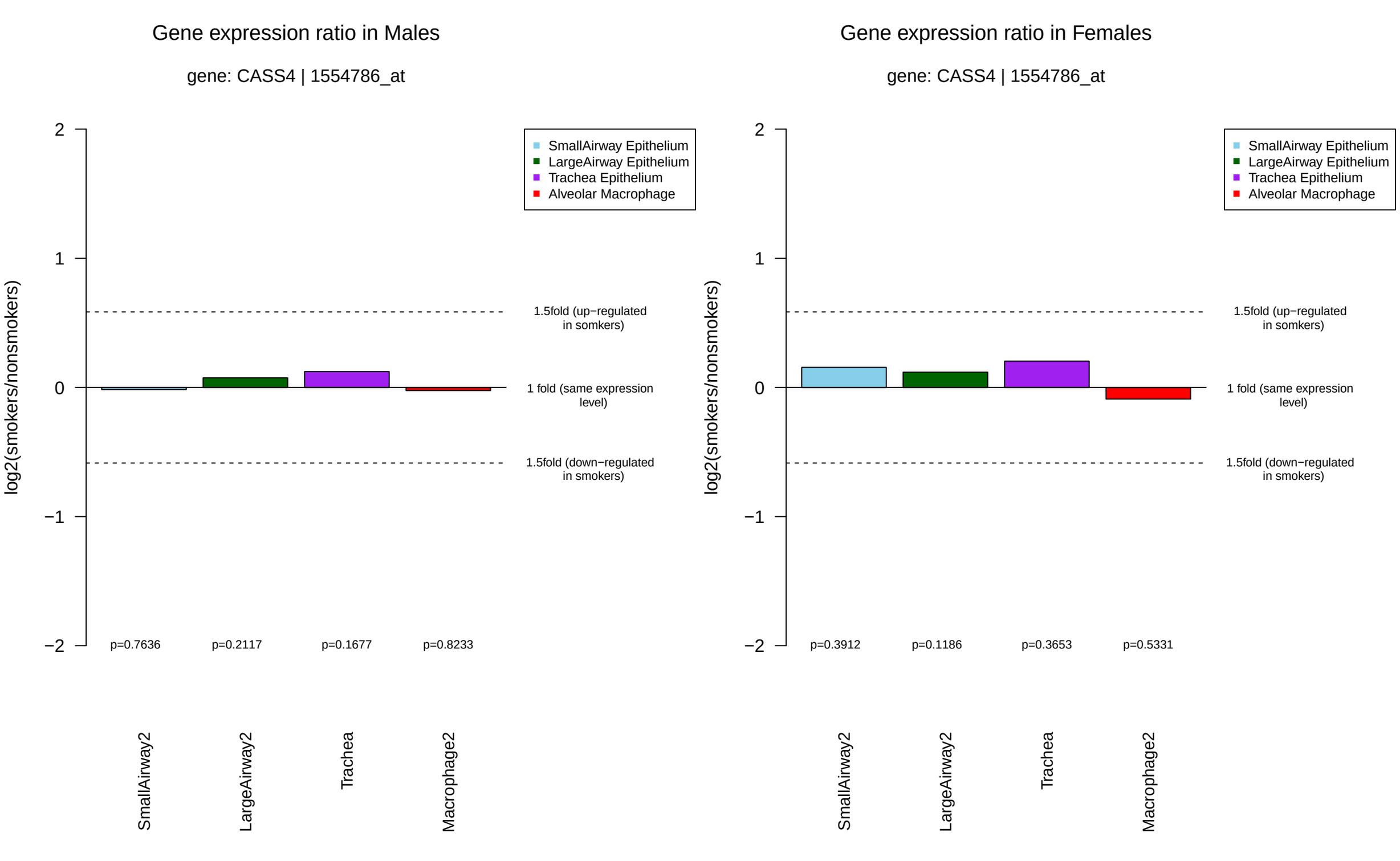
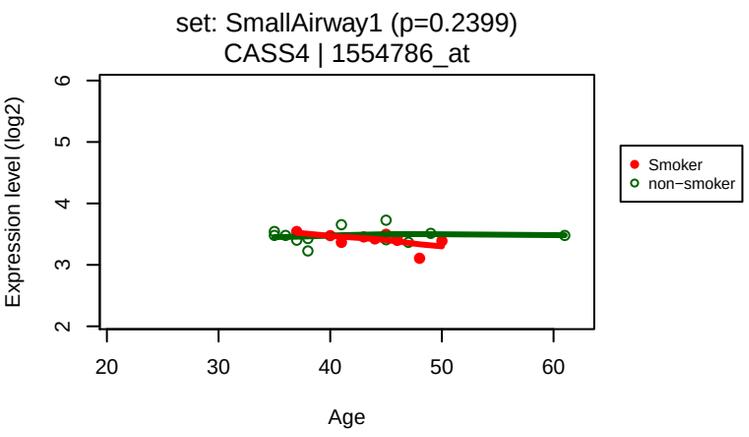
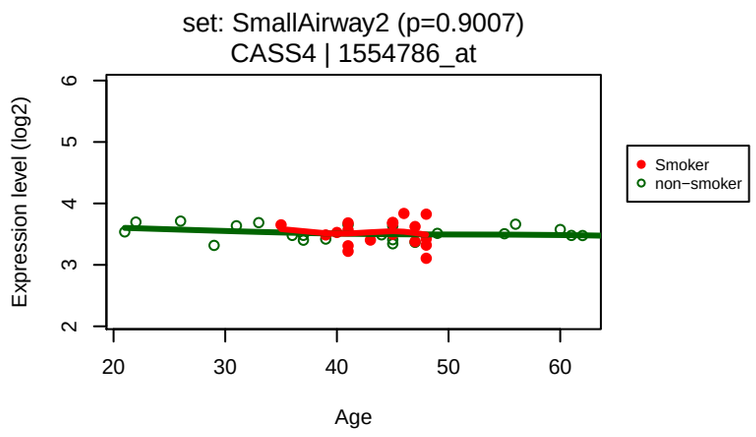
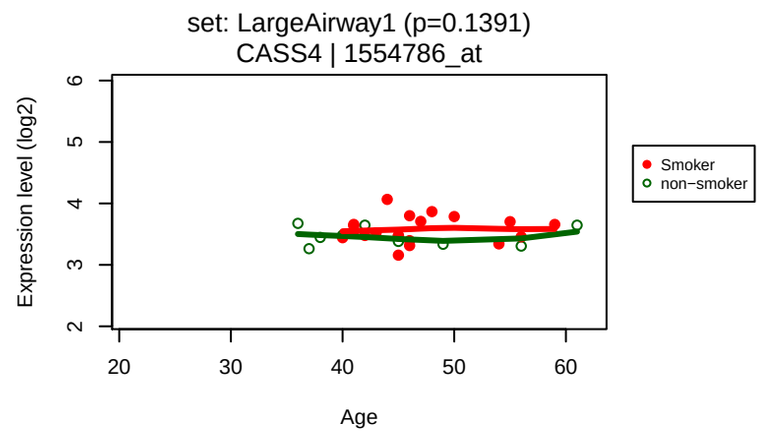

Gene expression of smokers and nonsmokers
gene: CASS4 | 1554786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2399
p=0.9007
p=0.8988
p=0.1391
p=0.0294
p=0.0925
p=0.6499
p=0.5734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

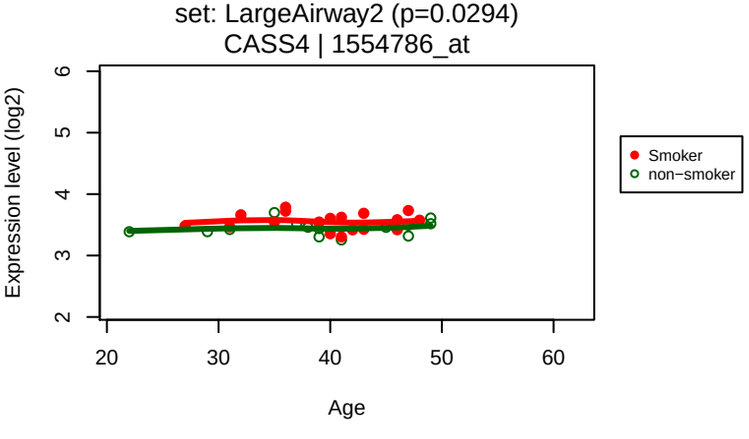

Gene expression ratio (smokers/nonsmokers)
gene: CASS4 | 1554786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2399
p=0.9007
p=0.8988
p=0.1391
p=0.0294
p=0.0925
p=0.6499
p=0.5734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

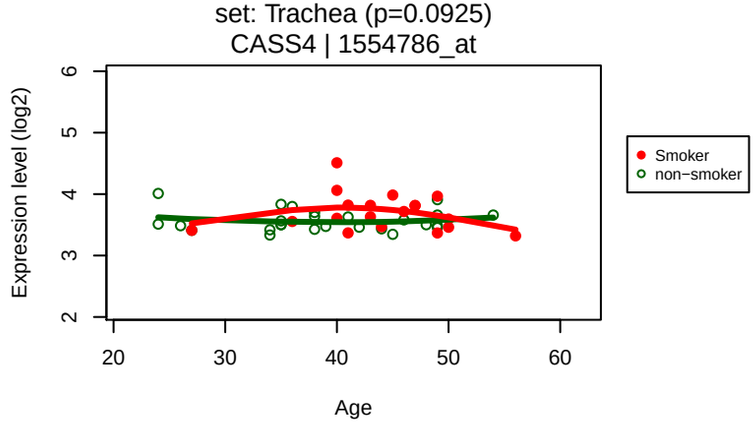

Gene expression ratio in Males
gene: CASS4 | 1554786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7636
p=0.2117
p=0.1677
p=0.8233
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASS4 | 1554786_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3912
p=0.1186
p=0.3653
p=0.5331
SmallAirway2
LargeAirway2
Trachea
Macrophage2

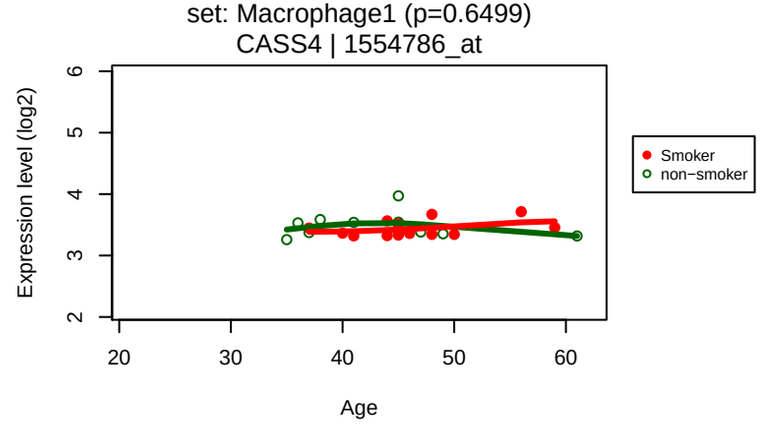

set: SmallAirway1 (p=0.2399)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker

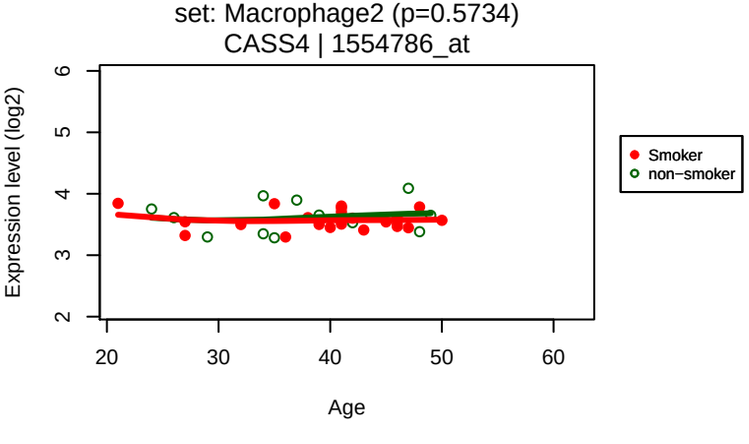

set: SmallAirway2 (p=0.9007)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker

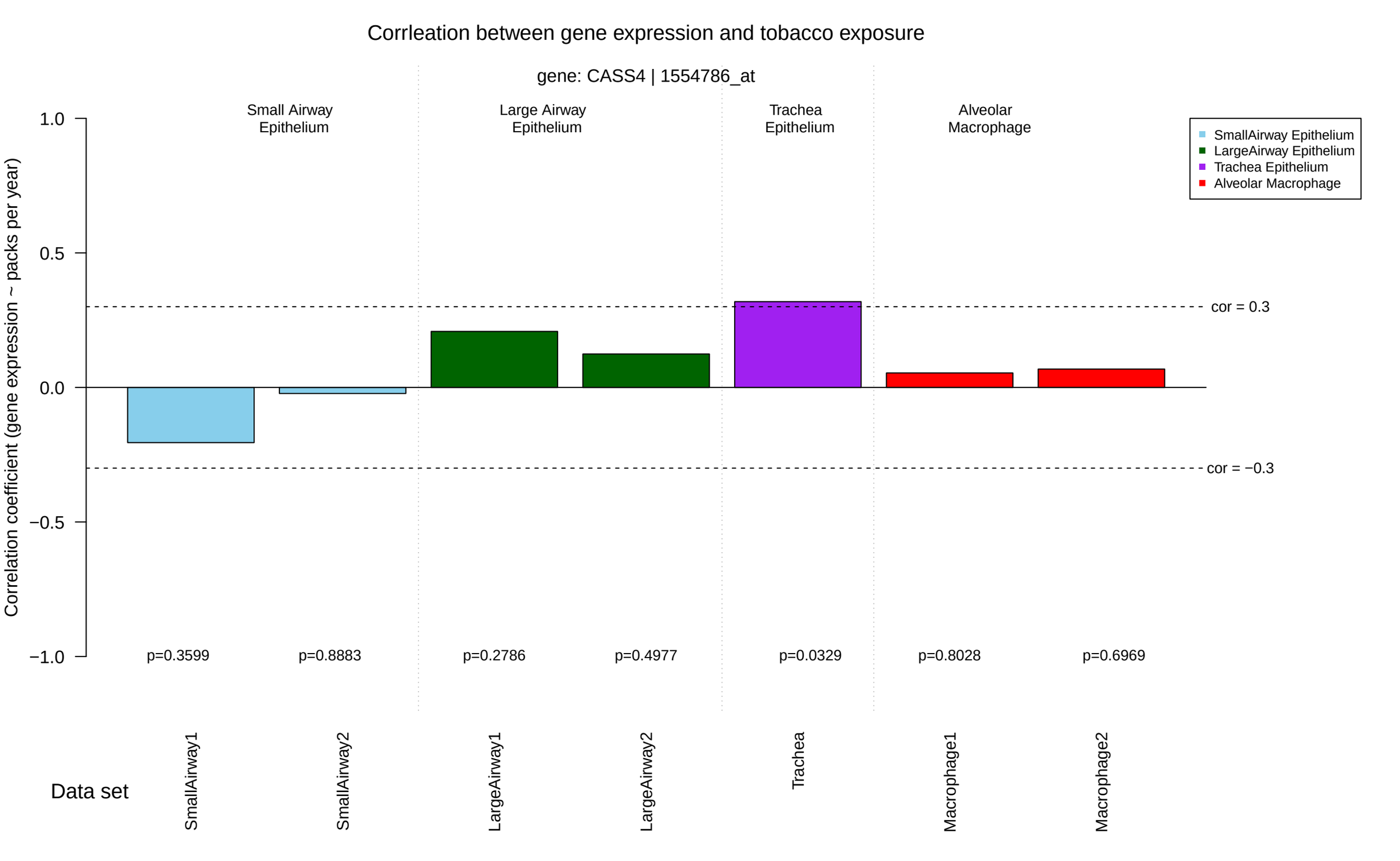

set: LargeAirway1 (p=0.1391)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0294)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0925)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6499)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5734)
CASS4 | 1554786_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASS4 | 1554786_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3599
p=0.8883
p=0.2786
p=0.4977
p=0.0329
p=0.8028
p=0.6969
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2